Graphic Era Hill University, Bhimtal

Department of Computer Science and Engineering

B.Tech. CSE - Semester III - Academic Year 2026-27 (Odd Semester)
For *Sections B & C*

Assignment 1 - Unit I

TCS302: Data Structures with C

Introduction, Arrays, Stacks and Recursion

Course Outcome	Maximum Marks	Faculty	Due Date
CO1	20	Mr. Manoj Kaushik	8 Aug 2026

Instructions: *Attempt all ten questions. Each question carries 2 marks. Write all C code by hand or in your lab notebook - code must compile as written. Show the working (traces, stack states) wherever asked; the trace carries the marks, not just the final answer.*

Note: Submit the physical copy of the assignment.

Q1. *[Theory]*

Define a data structure. Distinguish between linear and non-linear data structures, giving two examples of each.

Q2. *[Theory]*

What do time complexity and space complexity measure? State the time complexity, using Big-O notation, of: (a) searching an unsorted array of n elements, (b) push on a stack, (c) accessing arr[i] in an array, (d) displaying all n elements of a stack.

Q3. *[Numerical]*

An integer array A[10][20] is stored in row-major order at base address 2000, with each element occupying 4 bytes. Calculate the address of A[4][7]. Show the formula and each substitution step.

Q4. *[Coding]*

Write a C function `int findMax(int arr[], int n)` that returns the largest element of an array of n integers. State its time complexity in one line below the code.

Q5. *[Theory]*

For an array stack with capacity MAX, state the exact conditions for stack overflow and stack underflow, and name the operation in which each must be checked. Why does the empty stack use top = -1 rather than top = 0?

Q6. *[Coding]*

Using the declarations `int stack[MAX]; int top = -1;` write the C functions push() and pop() for an array stack, including the overflow and underflow checks from Q5.

Q7. *[Theory + Trace]*

In a linked stack, why must the statement `node->next = s->top;` come before `s->top = node;` inside push? Explain, with a small diagram, exactly what is lost if the two lines are swapped.

Q8. *[Trace]*

Convert the infix expression (A + B) * C - D / E to (a) postfix and (b) prefix. For the postfix conversion, show the operator stack and the output after every symbol is read.

Q9. *[Trace]*

Evaluate the postfix expression 8 2 / 3 + 6 2 * - showing the operand stack after every symbol. In the operator step, clearly label which popped value is left and which is right.

Q10. *[Coding + Theory]*

Write a recursive C function `int sumDigits(int n)` that returns the sum of the digits of a positive integer (for example, sumDigits(493) = 16). Clearly mark the base case and the recursive case as comments, and state how many stack frames are created for a 5-digit number.

Answers to numerical and trace questions will be discussed in the classroom after submission.